Formulas Reference: DDPM, DDIM, DPS, NDTM

1 DDPM (Denoising Diffusion Probabilistic Models)

1.1 Forward process

The forward process progressively adds noise to the data:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t;\ \sqrt{\alpha_t}\,\mathbf{x}_{t-1},\ \beta_t\mathbf{I}) \tag{1}$$

where $\alpha_t = 1 - \beta_t$ and β_t follows a linear schedule from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$.

Thanks to the Markov structure, we can sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}\big) \tag{2}$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{3}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$.

1.2 Tweedie estimate (prediction of $\mathbf{x}_0$ from $\mathbf{x}_t$)

$$\hat{\mathbf{x}}_0(\mathbf{x}_t) = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) \tag{4}$$

1.3 Reverse process (DDPM sampling)

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}\Big(\mathbf{x}_{t-1};\ \boldsymbol{\mu}_\theta(\mathbf{x}_t, t),\ \tilde{\beta}_t\mathbf{I}\Big) \tag{5}$$

where the posterior mean is:

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) \tag{6}$$

and the posterior variance is:

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,\beta_t \tag{7}$$

The sampling step is:

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sqrt{\tilde{\beta}_t}\,\mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{8}$$

1.4 Training objective

$$\mathcal{L} = \mathbb{E}_{t,\,\mathbf{x}_0,\,\boldsymbol{\epsilon}}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2\right] \tag{9}$$

2 DDIM (Denoising Diffusion Implicit Models)

DDIM defines a non-Markovian reverse process that shares the same marginals $q(\mathbf{x}_t \mid \mathbf{x}_0)$ as DDPM but allows deterministic sampling with fewer steps.

2.1 Generalized reverse step

Given a subsequence of timesteps $\tau_1 < \tau_2 < \cdots < \tau_S \subseteq \{1, \ldots, T\}$, the DDIM update from $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$ (where t and $t-1$ refer to consecutive elements in the subsequence) is:

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \underbrace{\hat{\mathbf{x}}_0(\mathbf{x}_t)}_{\text{predicted } \mathbf{x}_0} + \underbrace{\sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\, \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}_{\text{direction pointing to } \mathbf{x}_t} + \underbrace{\sigma_t\, \mathbf{z}}_{\text{noise}} \tag{10}$$

where:

$$\hat{\mathbf{x}}_0(\mathbf{x}_t) = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\, \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{11}$$

The noise scale σ_t interpolates between DDPM and deterministic:

$$\sigma_t = \eta \sqrt{\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}} \sqrt{1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}} \tag{12}$$

- $\eta = 0$: fully deterministic (DDIM)
- $\eta = 1$: recovers DDPM stochasticity

2.2 Key properties

- Uses the **same pretrained model** $\boldsymbol{\epsilon}_\theta$ as DDPM — no retraining.
- Can use far **fewer steps** (e.g. 50 instead of 1000) by subsampling the timestep schedule.
- The $\bar{\alpha}_t$ values are looked up from the **original** $T = 1000$ schedule at the selected subsequence indices.

3 DPS (Diffusion Posterior Sampling)

3.1 Problem setup

Given a degradation model:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \sigma_y\, \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{13}$$

we want to sample from the posterior $p(\mathbf{x}_0 \mid \mathbf{y})$.

3.2 Conditional score decomposition

Using Bayes' rule on the score:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t \mid \mathbf{y}) = \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)}_{\text{unconditional score}} + \underbrace{\nabla_{\mathbf{x}_t} \log p(\mathbf{y} \mid \mathbf{x}_t)}_{\text{likelihood score}} \tag{14}$$

3.3 DPS approximation

The intractable likelihood $p(\mathbf{y} \mid \mathbf{x}_t) = \int p(\mathbf{y} \mid \mathbf{x}_0)\, p(\mathbf{x}_0 \mid \mathbf{x}_t)\, d\mathbf{x}_0$ is approximated using a Dirac delta at the Tweedie estimate:

$$p(\mathbf{x}_0 \mid \mathbf{x}_t) \approx \delta(\mathbf{x}_0 - \hat{\mathbf{x}}_0(\mathbf{x}_t)) \tag{15}$$

This gives:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y} \mid \mathbf{x}_t) \approx -\nabla_{\mathbf{x}_t} \left\| \mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_0(\mathbf{x}_t)) \right\|^2 \tag{16}$$

where the gradient flows through the Tweedie estimate $\hat{\mathbf{x}}_0$ back to $\mathbf{x}_t$ via the chain rule.

3.4 DPS sampling step

Combining the unconditional DDIM step with the guidance gradient:

$$\mathbf{x}_{t-1} = \underbrace{\sqrt{\bar{\alpha}_{t-1}}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t,t) + \sigma_t\mathbf{z}}_{\text{standard DDIM step}} - \underbrace{\zeta_t\,\frac{\nabla_{\mathbf{x}_t}\left\|\mathbf{y}-\mathcal{A}(\hat{\mathbf{x}}_0)\right\|^2}{\left\|\nabla_{\mathbf{x}_t}\left\|\mathbf{y}-\mathcal{A}(\hat{\mathbf{x}}_0)\right\|^2\right\|}}_{\text{DPS guidance (norm-normalized)}} \tag{17}$$

where ζ_t is the guidance step size.

3.5 Terminal costs for specific tasks

For **deblurring** with blur kernel $\mathbf{k}$:

$$\Phi(\hat{\mathbf{x}}_0) = \|\mathbf{k} * \hat{\mathbf{x}}_0 - \mathbf{y}\|^2 \tag{18}$$

For **inpainting** with binary mask $\mathbf{M}$:

$$\Phi(\hat{\mathbf{x}}_0) = \|\mathbf{M} \odot \hat{\mathbf{x}}_0 - \mathbf{y}\|^2 \tag{19}$$

For **non-linear deblurring** with operator $\mathcal{A}$ (e.g. blur + bicubic resize):

$$\Phi(\hat{\mathbf{x}}_0) = \|\mathcal{A}(\hat{\mathbf{x}}_0) - \mathbf{y}\|^2 \tag{20}$$

4 NDTM (Non-linear Diffusion Trajectory Matching)

4.1 Variational control formulation

At each timestep t, NDTM introduces a control signal $\mathbf{u}_t$ and optimizes:

$$\mathcal{C}(\mathbf{u}_t) = w_T\,\Phi(\hat{\mathbf{x}}_0^{(t)}) + D_{\mathrm{KL}}(p(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{u}_t) \parallel q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)) \tag{21}$$

where Φ is the terminal cost and the KL term keeps the guided trajectory close to the unconditional one.

4.2 NDTM objective (DDIM instantiation, Proposition 3.1)

For DDIM, the objective simplifies to:

$$\mathcal{C}(\mathbf{u}_t) = \underbrace{\kappa_t^2\,\|\mathbf{u}_t\|^2}_{\text{control cost}} + \underbrace{\tau_t^2\,\|\boldsymbol{\epsilon}_\theta(\bar{\mathbf{x}}_t,t) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t,t)\|^2}_{\text{score cost}} + \underbrace{w_T\,\Phi(\hat{\mathbf{x}}_0^{(t)})}_{\text{terminal cost}} \tag{22}$$

where $\bar{\mathbf{x}}_t = \mathbf{x}_t + \gamma\,\mathbf{u}_t$ is the guided state, and the coefficients are:

$$\tau_t = \sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2} - \sqrt{\frac{\bar{\alpha}_{t-1}(1-\alpha_t)}{\alpha_t}}, \qquad \kappa_t = \gamma\,\frac{\sqrt{\bar{\alpha}_{t-1}}}{\sqrt{\alpha_t}} \tag{23}$$

4.3 NDTM sampling algorithm

At each diffusion step t:

1. Initialize $\mathbf{u}_t^{(0)} = \mathbf{0}$
2. Compute unconditioned prediction: $\hat{\boldsymbol{\epsilon}}_{\text{uncond}} \leftarrow \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$
3. For $i = 0, \ldots, N-1$ (inner optimization):

(a) Compute controlled prediction: $\hat{\boldsymbol{\epsilon}}_{\text{ctrl}}^{(i)} \leftarrow \boldsymbol{\epsilon}_\theta(\mathbf{x}_t + \gamma\, \mathbf{u}_t^{(i)}, t)$

(b) Compute Tweedie estimate: $\hat{\mathbf{x}}_0^{(i)} \leftarrow \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t + \gamma\, \mathbf{u}_t^{(i)}]$

(c) Compute total cost $\mathcal{C}^{(i)}$ and update: $\mathbf{u}_t^{(i+1)} \leftarrow \mathbf{u}_t^{(i)} - \eta_{\text{lr}}\, \nabla_{\mathbf{u}_t} \mathcal{C}^{(i)}$

4. Sample from guided posterior: $\mathbf{x}_{t-1} \leftarrow \text{DDIM}(\mathbf{x}_t + \gamma\, \mathbf{u}_t^*, t)$

4.4 Comparison with DPS

DPS can be seen as a special case of the DTM framework with linear control ($\mathbf{u}_t$ added directly to the score) and no transient cost regularization. Specifically, with $\kappa_t = 0$, $\tau_t = 0$, and $\gamma = 1$, the NDTM update reduces to a single gradient step on the terminal cost, recovering the DPS guidance term.

5 Summary of relationships

	DDPM	**DDIM**	**DPS**	**NDTM**
Stochastic	Yes ($\tilde{\beta}_t$)	Controlled by η	Inherits from DDIM	Inherits from DDIM
Steps needed	$T = 1000$	$S \ll T$ (e.g. 50)	S (100–1000)	S (e.g. 50)
Guidance	None	None	$-\zeta_t \nabla_{\mathbf{x}_t} \Phi(\hat{\mathbf{x}}_0)$	$\min_{\mathbf{u}_t} \mathcal{C}(\mathbf{u}_t)$
Inner loop	No	No	No	Yes (N steps)
Model	$\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$	Same	Same	Same
Markovian	Yes	No	No	No